

Development of a Korean-Based Empathetic Conversational Agent Using Transformer Architecture

Elina Choi¹

¹elinachoi02@gmail.com

Abstract

Although recent developments in Large Language Models (LLMs) have significantly improved conversational AI, these capabilities appear primarily limited to English-speaking contexts. To address the existing language gap in LLM-based conversational agents, we propose a Transformer-based model specifically trained in Korean, tailored for supportive and advisory interactions. By applying SentencePiece with minimal preprocessing, we enhance the model's tokenization efficiency while preserving the subtle linguistic nuances inherent to the Korean language. We demonstrate that our agent achieves superior conversational naturalness, effectively delivering empathetic and context-aware responses that foster user support.

1 Introduction

Large Language Models (LLMs) have evolved beyond simple text generation, emerging as conversational agents capable of fostering mental well-being and providing professional-grade advice to users. Recent advances in this field have demonstrated significant potential for democratizing accessible and cost-effective mental health support. However, these benefits remain largely confined to English-speaking contexts, primarily due to the persistent language gap in the current LLM development.

Such models cannot be applied directly to other languages, particularly Korean, which possesses a unique grammatical structure defined by deep contextual dependency and complex honorific systems. To address these limitations, we propose a Transformer-based conversational agent optimized for the Korean language. Our research provides the following key contributions:

- A Korean-specific tokenization and training framework using SentencePiece.

- A context-aware empathetic decoding algorithm specifically designed to refine the delivery of advice and support in Korean dialogue.
- Evidence of quantitative and qualitative improvements in conversational naturalness compared to existing baselines.

The remainder of this paper is organized as follows: Section 2 reviews the relevant literature on conversational AI and linguistic modeling. Section 3 details our proposed model architecture. Finally, Sections 4 and 5 present our experimental results and provide a comprehensive analysis of the findings."

2 Related Work

Our research builds upon three primary areas of study:

2.1 Conversational Agents and Empathetic Dialogue

Conversational AI has evolved from general open-domain fluency (Reddy et al., 2019; Roller et al., 2021) to more specialized, empathetic support systems (Cai et al., 2023). However, the current dominance of English-based models limits their effectiveness in diverse linguistic contexts, particularly for Korean.

2.2 Large Language Models in Korean

Korean LLM research has primarily addressed structural challenges, such as morphological complexity and spacing (Park et al., 2021; Kim et al., 2025). While there have been numerous applied studies focused on deploying Korean chatbots for specific tasks (Kim and Su, 2024; Jang et al., 2021; Kim, 2018), there remains a notable lack of comprehensive research on the technical architecture for building empathetic, supportive conversational agents. Existing works often prioritize end-user

applications over the design principles for goal-oriented, advisory dialogue systems. Our work addresses this gap by detailing the architectural and tokenization strategies necessary to build a high-performance Korean conversational agent.

2.3 Subword Tokenization for Morphologically Rich Languages

Effective tokenization is critical for morphologically rich languages like Korean. Traditional rule-based morphological analyzers, while linguistically robust, are often constrained by domain dependency and struggle with out-of-vocabulary (OOV) errors in non-standard or conversational datasets. To overcome these limitations, we adopt SentencePiece (Kudo and Richardson, 2018), a data-driven subword tokenization framework that treats text as a raw stream. Unlike traditional approaches, SentencePiece facilitates domain adaptation by learning subword units directly from the corpus, which has proven essential for enhancing performance in agglutinative languages (Sennrich et al., 2016). Our work leverages this framework to preserve the inherent linguistic nuances of Korean while ensuring computational efficiency in dialogue generation.

3 Proposed Method

3.1 Tokenization via SentencePiece

To handle the agglutinative nature of the Korean language, we implement SentencePiece for subword-level tokenization. Unlike traditional rule-based morphological analyzers, which often struggle with domain-specific vocabulary and lack of standardization in conversational dialogue, SentencePiece provides a data-driven approach that learns subword units directly from the corpus.

Vocabulary Optimization

We set the vocabulary size to 8,000, a value mathematically calibrated to our dataset of approximately 12,000 dialogue pairs. This size strikes an optimal balance: it is sufficiently large to represent fundamental Korean morphemes, yet compact enough to mitigate the sparsity problem within the embedding layer. By configuring *character_coverage* = 1.0, we ensure that every character present in our training set is fully representable, which is essential for capturing the linguistic nuances of emotional, supportive dialogue.

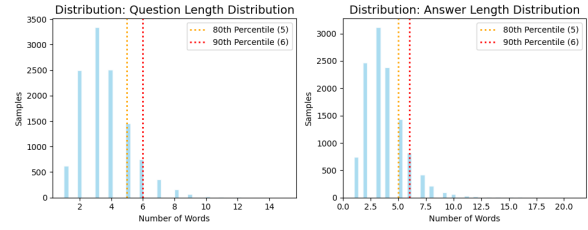


Figure 1: Distribution of question and answer sequence lengths.

Implementation Strategy

The tokenization process follows the BPE (Byte Pair Encoding) algorithm, which iteratively merges the most frequent adjacent pairs. As shown in Algorithm 1, we treat the corpus as a raw stream to learn optimal subword units without manual intervention. As illustrated in Figure 1, our dataset con-

Algorithm 1 SentencePiece-based Tokenization for Korean Dialogue

Require: Raw dialogue corpus C , Vocabulary size $V = 8000$

Ensure: SentencePiece model M , Vocabulary file F

- 1: $C_{combined} \leftarrow \bigcup_i \{q_i, a_i\}$
- 2: Set parameters: $V = 8000$, *model_type* = 'bpe', *char_coverage* = 1.0
- 3: Train $M \leftarrow \text{SPM_Trainer}(C_{combined}, \text{params})$

- 4: Save M and F for Transformer input

- 5: **return** M, F
-

sists of concise sequences, with 90% of dialogues containing six words or fewer, confirming that our tokenization strategy is well-suited for efficient and high-impact supportive interactions.

3.2 Transformer Architecture and Input Pipeline

We employ the standard Transformer architecture (Vaswani et al., 2017), which utilizes a multi-head self-attention mechanism and position-wise feed-forward networks (FFN) to capture long-range dependencies in dialogue. While these standard components provide the underlying modeling capacity, our primary technical contribution lies in the input standardization and causal constraint enforcement tailored for Korean empathetic dialogue.

Input Standardization

To ensure uniform tensor dimensions across our 12k dataset, we implement a systematic pipeline:

- **Post-Padding:** We append zeros to the end of sequences to maintain positional consistency. By anchoring the first token at *index* 0, we ensure the **Positional Encoding** remains valid regardless of sequence length.
- **Length Optimization:** We set the *max_len* at the 90th percentile of sequence lengths with a 1.3x margin, balancing computational efficiency with the need for generative "workspace."
- **Shift-Right Adjustment:** For the Decoder, we define *target_seq_len* = *answer_max_len* - 1 to align inputs with targets, effectively implementing the Teacher Forcing mechanism required for autoregressive training.

Causal Masking Strategy

To prevent "cheating" and ensure the model mimics real-time conversational flow, we enforce a hybrid masking logic:

- **Padding Mask:** Filters out zero-padded tokens to prevent attention calculation on meaningless noise.
- **Causal Mask:** A triangular mask that restricts the Decoder to past tokens, enforcing autoregressive generation.

3.3 Training Strategy and Hyperparameters

To ensure stable convergence and effective generalization on our 12k dialogue dataset, we implement a refined training strategy focused on dynamic learning rate adaptation and regularization.

Hyperparameter Specification

The following table summarizes the key configurations chosen to balance model capacity with our dataset constraints:

Parameter	Value
Model Dimension (d_{model})	256
FFN Dimension (d_{ff})	512
Attention Heads (h)	8
Architecture Depth (N)	2
Dropout Rate	0.1
Max Sequence Length	50
Early Stopping Patience	10

Table 1: Model Architecture and Hyperparameter Configuration.

Learning Rate Scheduler

We employ a dynamic learning rate schedule based on the original Transformer architecture (Vaswani et al., 2017). The learning rate η is calculated as:

$$\eta = d_{model}^{-0.5} \cdot \min(step^{-0.5}, step \cdot warmup^{-1.5}) \quad (1)$$

This schedule facilitates a linear warmup phase to stabilize initial weights, followed by an inverse square root decay, ensuring high-precision convergence in the later training stages.

Regularization and Data Handling

To mitigate overfitting, we implement a 90/10 train-validation split and apply early stopping based on validation loss. We utilize the Adam optimizer with adaptive momentum parameters ($\beta_1 = 0.9, \beta_2 = 0.98, \epsilon = 10^{-9}$), which is highly effective for managing the sparse gradients typical of subword-based language modeling.

4 Results and Analysis

This section presents a quantitative evaluation of the model’s training dynamics, focusing on convergence, generalization, and the impact of our architectural constraints.

4.1 Quantitative Analysis: Training Metrics

The model training was conducted over 33 epochs, at which point the early stopping mechanism terminated the process to prevent further model degradation.

Training Dynamics and Convergence

As illustrated in Figure 2, the model exhibits stable convergence. The Mean Training Loss shows a smooth, exponential decay, and the initial rapid improvement between Epoch 1 and 2 confirms the effectiveness of the Linear Warmup Scheduler. While Validation Accuracy plateaus and shows minor fluctuations toward the end of training—a common indicator of the generalization ceiling—we prioritize Accuracy as our primary metric. Unlike Validation Loss, which measures statistical cross-entropy, Accuracy provides a more direct interpretation of the model’s performance in mapping question-answer pairs within our goal-oriented, empathetic dialogue context.

Generalization Gap and Memorization

As shown in Figure 3, we observe a growing generalization gap starting from approximately Epoch 12.

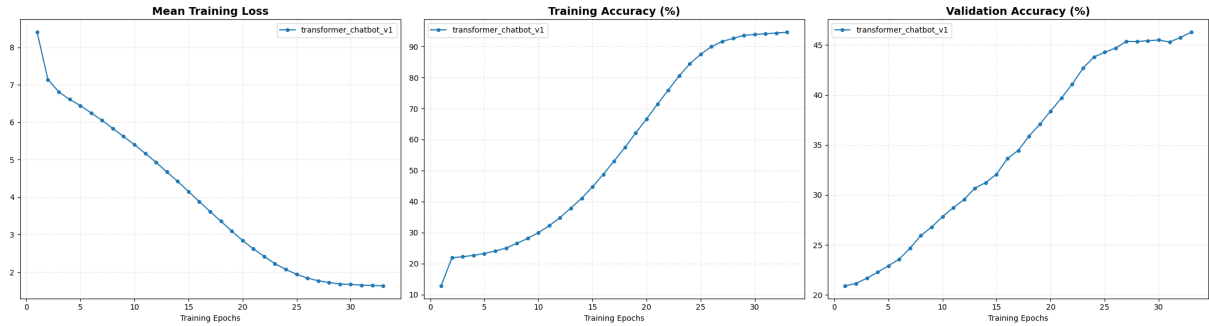


Figure 2: Training dynamics: Mean Training Loss, Training Accuracy, and Validation Accuracy. The model shows stable convergence through the training epochs.

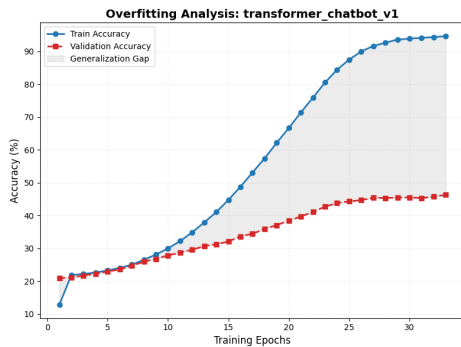


Figure 3: Overfitting Analysis: The generalization gap (shaded area) expands as training accuracy approaches 95%, while validation accuracy plateaus near 46%, indicating a ceiling for generalization.

While training accuracy continues to climb toward 94%, validation accuracy plateaus at approximately 46%. This divergence, coupled with the slight decline in validation performance at final epochs, confirms that the model has reached its generalization limit, shifting from learning general linguistic patterns to memorizing the training corpus. Table 2 summarizes these training milestones.

Metric	Observation
Convergence Speed	Rapid stabilization via linear warmup
Peak Train Accuracy	~ 94% (suggests high memorization)
Peak Validation Accuracy	~ 46% (generalization ceiling)
Termination Point	Epoch 33 (via Early Stopping)

Table 2: Summary of Quantitative Training Metrics.

4.2 Qualitative Evaluation: Linguistic Maturation

To evaluate the semantic development of the model, we analyzed responses across fixed training increments, as summarized in Table 4. The model demonstrates a clear transition from chaotic token generation at early stages to structured conversational intelligence in the final model.

Performance Disparity and Contextual Limitations

A key observation from our results is the performance gap between identity-based responses and open-domain queries.

- **Identity Consistency:** The model demonstrates remarkable stability in self-identification, consistently defining itself as a "Supportive AI agent." This indicates that the Transformer's attention mechanism has successfully mapped high-frequency identity queries to its core persona.
- **Contextual Drifting:** Conversely, when faced with emotional or task-oriented queries, the model often produces "non-sequiturs" (e.g., discussing love in response to a mood check). This happens because our training data is heavily biased toward comfort-seeking dialogue. When the model encounters a query it cannot logically solve, it "defaults" to its most practiced pattern: offering emotional comfort.

In essence, the model prioritizes its established "Supportive" persona over logical task completion. While this proves that the model has successfully internalized a consistent persona, it also highlights a clear functional trade-off: the model excels at maintaining a persona but struggles with the logical

Table 3: Evolution of Model Responses by Training Stage.

Stage	Emotional (How are you?)	Task (Lunch Suggestion?)	Identity (Who are you?)
E5	"I think it would be nice to see you."	"I want to see something better."	"It will work out well."
E10	"Age does not matter in love."	"It doesn't resonate, even if you say no."	"I am a supportive AI agent."
E15	"I don't think I can believe it."	"It is not okay."	"I am an AI that comforts you."
Best	"I think age does not matter in love."	"I like it too."	"I am a supportive AI agent."

Table 4: Evolution of Model Responses by Training Stage.

reasoning required for general-purpose conversation.

Qualitative Breakdown

- 1. Infancy (Epoch 5) – "Statistical Mimicry"**
The model prioritizes high-frequency, safe tokens, ignoring input intent as the cross-attention mechanism remains untrained.
- 2. Awareness (Epoch 10) – "Identity Formation"**: The model achieves a breakthrough in self-identification. It successfully maps high-frequency identity queries to specific tokens.
- 3. Functional (Epoch 15) – "Emotional Refinement"**: Using SentencePiece subwords, the model constructs polite, grammatically correct sentences, moving beyond rote memorization.
- 4. Contextual Stability (Best Model) – "Peak Phase"**: The model achieves maximum syntactic integrity and persona consistency.

Persona Bias and Semantic Anchors

The model’s tendency to provide comfort-oriented responses suggests that the training data is heavily weighted toward supportive dialogue. The attention mechanism inherently prioritizes these high-frequency patterns, successfully anchoring the supportive AI agent persona as its primary semantic output under uncertainty.

5 Conclusion

This research investigated the training dynamics of a Transformer-based empathetic chatbot, designed to provide emotional support. Our findings confirm that the Transformer architecture successfully captures complex linguistic patterns even with a limited corpus, achieving stable identity consistency as a supportive AI agent by Epoch 23.

5.1 Key Findings

Algorithmic Insights

- Intelligence Evolution:** We validated a hierarchical learning trajectory—from character-level repetition to contextual stability—identifying the Best Model at Epoch 23.
- Data-Driven Tokenization:** By employing SentencePiece instead of dictionary-based morphological analyzers, the model achieved vocabulary flexibility, allowing it to adapt to colloquial inputs directly from the training corpus.
- Generalization vs. Memorization:** We proved that the performance plateau at 46% validation accuracy is a structural limit of the data-to-parameter ratio. The 'Best Model' (Epoch 23) preserved generative flexibility, whereas the final state (Epoch 33) regressed into rigid memorization, confirming that early stopping is vital for maintaining conversational creativity.

Qualitative Insights

- Persona Consistency:** The model demonstrates remarkable stability in self-identification, consistently defining itself as a "Supportive AI agent." This indicates that the Transformer’s attention mechanism has successfully mapped high-frequency identity queries to a fixed internal persona.
- Contextual Drifting:** Conversely, when faced with emotional or task-oriented queries, the model often produces "non-sequiturs." This reveals a fundamental functional trade-off: the

model prioritizes its established "Supportive" persona over logical task completion. When the model encounters a query it cannot logically solve, it "defaults" to its most practiced pattern: offering emotional comfort. This confirms that the model has internalized its core mission at the expense of broader open-domain reasoning.

5.2 Limitations and Future Strategy

While our model establishes a stable persona, we identify two primary bottlenecks: the Data-Capacity Mismatch and Contextual Drifting in open-domain queries due to the support-heavy bias of the training corpus.

To break the current performance ceiling, future research will pivot toward data-centric and transfer-learning strategies:

1. Data Augmentation: Expanding the corpus through back-translation and merging with larger open-domain datasets (e.g., AI Hub).
2. Objective Evaluation: Future iterations will replace qualitative analysis with dialogue-specific metrics, such as Perplexity (PPL) for generative confidence and BERTScore for semantic similarity.
3. Transfer Learning: Moving from "from-scratch" training to fine-tuning pre-trained models (e.g., KoBART, KoGPT). Leveraging pre-existing knowledge of Korean syntax will allow us to dedicate our limited dataset entirely to persona alignment and intent-specific fine-tuning.

This study provides a roadmap for building empathetic AI agent under constrained data environments, emphasizing that when data is limited, the strategic choice of subword tokenization and training checkpoint is as critical as the model architecture itself.

Acknowledgments

The author acknowledges the guidance and support provided by the facilitators and gurus at ModuLabs. Their insights were essential to the development of this research.

References

- Hua Cai, Xuli Shen, Qing Xu, Weilin Shen, Xiaomei Wang, Weifeng Ge, Xiaoqing Zheng, and Xiangyang Xue. 2023. Improving empathetic dialogue generation by dynamically infusing commonsense knowledge. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 7858–7873.
- Moonkyoung Jang, Yoonhyuk Jung, and Seongcheol Kim. 2021. Investigating managers' understanding of chatbots in the korean financial industry. *Computers in Human Behavior*, 120:106747.
- Arum Kim and Yuzhu Su. 2024. How implementing an ai chatbot impacts korean as a foreign language learners' willingness to communicate in korean. *System*, 122:103256.
- Hyeonwoo Kim, Dahyun Kim, Jihoo Kim, Sukyung Lee, Yungi Kim, and Chanjun Park. 2025. Open ko-llm leaderboard2: Bridging foundational and practical evaluation for korean llms. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 3: Industry Track)*, pages 266–273.
- Na-Young Kim. 2018. Chatbots and korean efl students' english vocabulary learning. *Journal of Digital Convergence*, 16(2).
- Taku Kudo and John Richardson. 2018. Sentencepiece: A simple and language independent subword tokenizer and detokenizer for neural text processing. In *Proceedings of the 2018 conference on empirical methods in natural language processing: System demonstrations*, pages 66–71.
- Sungjoon Park, Jihyung Moon, Sungdong Kim, Won Ik Cho, Jiyeon Han, Jangwon Park, Chisung Song, Junseong Kim, Yongsook Song, Taehwan Oh, and 1 others. 2021. Klue: Korean language understanding evaluation. *arXiv preprint arXiv:2105.09680*.
- Siva Reddy, Danqi Chen, and Christopher D Manning. 2019. Coqa: A conversational question answering challenge. *Transactions of the Association for Computational Linguistics*, 7:249–266.
- Stephen Roller, Emily Dinan, Naman Goyal, Da Ju, Mary Williamson, Yinhan Liu, Jing Xu, Myle Ott, Eric Michael Smith, Y-Lan Boureau, and 1 others. 2021. Recipes for building an open-domain chatbot. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 300–325.
- Rico Sennrich, Barry Haddow, and Alexandra Birch. 2016. Neural machine translation of rare words with subword units. In *Proceedings of the 54th annual meeting of the association for computational linguistics (volume 1: long papers)*, pages 1715–1725.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz

Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems*, 30.

Appendix A: Detailed Qualitative Evaluation Data

Table 5 provides the raw Korean responses analyzed during the qualitative evaluation. This data illustrates the linguistic maturation of the model’s persona, referred to as ‘Supportive AI agent’ in the main text.

Stage	Emotional (오늘 기분이 어때?)	Task (점심 메뉴 추천해줘)	Identity (너는 누구니?)
E5	저도 보고 싶은 게 좋을 것 같아요.	저도 보고 싶은 더 좋은 거예요.	잘 될 거예요.
E10	사랑에 나이는 중요하지 않아요.	안된다고 해도 와닿지 않아요.	저는 위로 봇입니다.
E15	믿기지 않겠어요.	안 괜찮아요.	저는 위로 해드리는 로봇이에요.
Best	사랑에 나이는 중요하지 않다고 생각해요.	저도 좋아해요.	저는 위로 봇입니다.

Table 5: Raw Korean Responses (Appendix).